

MENG WEI

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SUMMARY

AI Research Scientist & PhD Candidate (Available now, Smart Medical Imaging CDT) with extensive experience in AI for medical imaging and computer vision, specializing in deep learning methods. Expert in PyTorch and TensorFlow with strong mathematical foundations in linear algebra, calculus, and statistics. Demonstrated expertise in designing novel architectures, including vision-language models, graph neural networks, and transformers for complex medical imaging tasks. Skilled in data-efficient learning strategies. Strong publication record in top-tier venues demonstrating both technical innovation and clinical application. Experience working across diverse medical imaging modalities and collaborating with clinical teams to translate research into practical solutions. Strong communication skills for interdisciplinary audiences. Seeking a research scientist position to advance AI for medical imaging and clinical applications.

INTERNSHIP EXPERIENCE

Machine Learning Engineer Intern | Viridien (CGG), London, UK

Jun 2021 – Feb 2022

- Work in the 3D vision team on seismic images, leveraging advanced reconstruction techniques and Gaussian splatting to accurately model complex volumetric structures, demonstrating experience with Deep Learning techniques for production-ready solutions in an industry setting.
- Generated synthetic seismic data for deep learning model training, showcasing techniques for leveraging new data sources.
- Utilized NLP techniques to analyze textual files for estimating oil spot depths, aligning with preferences for NLP experience.
- Interacted with clients, providing expert suggestions to enhance traditional geoscientific models through AI techniques, demonstrating strong collaborative skills within a team.

Summer Research Intern | AI in Medical Image Lab, UCLA, USA

Jul 2019 – Sep 2019

- Proposed a novel deep learning framework for hemorrhagic stroke detection, incorporating symmetry constraints to enhance lesion contrast.
 - Introduced adversarial training into the network to improve segmentation accuracy and preserve lesion characteristics, optimizing the model for clinical stroke diagnosis.
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EDUCATION

PhD AI for Medical Imaging | King's College London

Oct 2021 – Sep 2025

UK Center of Doctoral Training - Awarded full 4-year funding for international doctoral study by the UK Engineering and Physical Sciences Research Council (EPSRC).

- Conducted fundamental and applied research in AI for medical imaging, focusing on developing new ML models and AI techniques for challenging real-world problems.
- Developed a semi-supervised learning framework leveraging large-scale unlabeled data for surgical instrument segmentation, demonstrating expertise in data-efficient model optimization.
- Contributed to a cutting-edge text-promptable approach for surgical instrument segmentation, integrating vision-language models, a mixture of text prompts, and a reinforcement module, showcasing capabilities in LLMs, NLP, and Multi-modal deep learning.
- Proposed a hybrid architecture combining pre-trained ViTs with CNNs via a lightweight adapter, demonstrating expertise in foundation model adaptation and transfer learning.
- Developed a part-aware instance segmentation framework with weakly-supervised learning on disjoint datasets, demonstrating expertise in multi-task learning under annotation constraints.
- Introduced a motion-guided framework for language-driven temporal segmentation in surgical videos with a multi-institutional benchmark dataset, showcasing skills in vision-language models, spatiotemporal reasoning, and video understanding.

Medical Robotics and Image-Guided Intervention pathway

Grade: Distinction (Outstanding Prize)

Core Modules: Computer vision, Machine learning, Deep learning, Python programming, Maths for machine learning, Advanced robotics, Complexity, Medical imaging.

Individual Project: Meta-learning for Small Lesion Detection

- Developed a meta-learning-based lesion detection model capable of accurately detecting small lesions with support from abundant large lesion data.
- Demonstrated significant improvements in detection precision for small lesions, contributing to advancements in CT images with potential clinical applications.

BEng Electrical Engineering | University of Liverpool**Sep 2017 – Jul 2020**

Grade: First-class honours

Core Modules: Signals and systems, Programming in C, Programming in C++, Communication systems, Engineering mathematics, Neural networks.

Final-year Project: CNN-GCN Aggregation Enabled Boundary Regression for Image Segmentation

- Developed an object detection model based on Fast R-CNN for localising discs and cups in retinal images, showcasing expertise in computer vision architectures.
- Proposed a novel CNN-GCN aggregation-enabled boundary regression model for image segmentation.

KEY PUBLICATION

- Wei, Meng, et al. "Where It Moves, It Matters: Referring Surgical Instrument Segmentation via Motion." In AAAI. 2026.
- Wei, Meng, et al. "Surgical Instances and Part Semantics for Weakly-supervised Part-aware Instance Segmentation." IEEE Transactions on Medical Imaging (2025): under rebuttal.
- Wei, Meng, et al. "SegMatch: semi-supervised surgical instrument segmentation." Scientific Reports 15.1 (2025): 14042.
- Wei, Meng, et al. "Enhancing surgical instrument segmentation: integrating vision transformer insights with adapter." International Journal of Computer Assisted Radiology and Surgery 19.7 (2024): 1313-1320.
- Zhou, Zijian, Wei, Meng, et al. "Text Promptable Surgical Instrument Segmentation with Vision-Language Models." In NeurIPS. 2023.
- Wei, Meng, et al. "Deep learning for hemorrhagic lesion detection and segmentation on brain CT images." IEEE journal of biomedical and health informatics 25.5 (2020): 1646-1659.
- Meng, Yanda, Wei, Meng, et al. "Regression of Instance Boundary by Aggregated CNN and GCN." In ECCV, pp. 190-207. 2020.

HONORS AND LEADERSHIP

- Dean's List honoree (Top 5% of Master's graduates), Imperial College London, 2020.
- Meritorious Winner of The Mathematical Contest in Modeling (MCM), 2018
- Vice President of Woman in Engineering Society, The University of Liverpool, 2018 - 2019

ADDITIONAL INFORMATION

- **Programming Languages:** Python (TensorFlow, PyTorch, Keras, OpenCV), Visual C, Visual C++, Verilog, Arduino; also basic ability with Java, R, HTML.
- **Industry Software Skills:** Matlab, Mathematics, SPSS.
- **Languages:** Mandarin (native), English (professional).

REFEREES

- **Tom Vercauteren**, Professor of Interventional Image Computing, King's College London. Email: tom.vercauteren@kcl.ac.uk
- **Stamatia Giannarou**, Senior Lecturer, Department of Surgery & Cancer, Imperial College London. Email: stamatia.giannarou@imperial.ac.uk
- **Miaoqing Shi**, Professor at Tongji University. Email: mshi@tongji.edu.cn